Source Data — INARA Project

CS6140 Machine Learning · Spring 2026 · Northeastern University San Jose
Wankhare, Shinde, Mulani | Dataset: INARA ATMOS (NASA FDL / Zorzan et al. 2025)

1. Scientific Motivation & Context

Determining the chemical composition of a distant planet's atmosphere is one of the central goals of exoplanet science. When a planet passes in front of its star (*transits*), starlight filters through the atmosphere. Different molecules absorb specific wavelengths, leaving distinctive fingerprints in the transmitted spectrum. Inferring which molecules are present — and in what quantities — is called **atmospheric retrieval**.

Traditional Bayesian retrieval (e.g. nested sampling, MCMC) can take **hours to days per planet**. The James Webb Space Telescope (JWST) and the planned Habitable Worlds Observatory (HWO) will observe thousands of exoplanet atmospheres. Fast, automated ML retrieval running in **milliseconds** is a scientific necessity for population-level surveys.

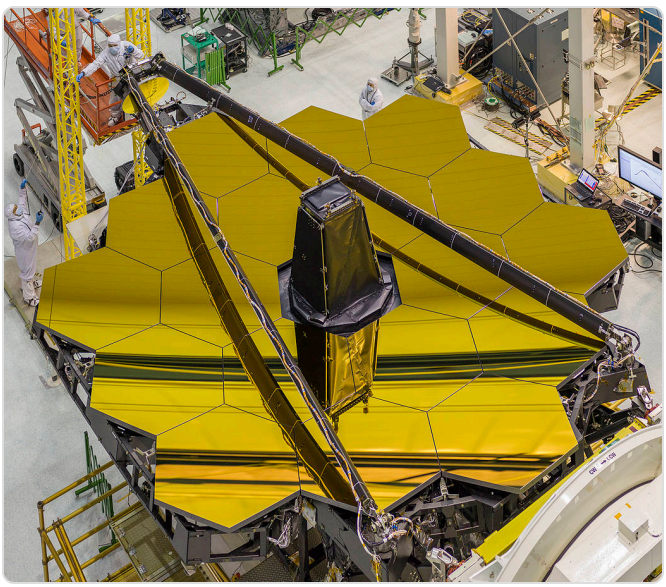

Fig 1. The James Webb Space Telescope with its iconic 6.5 m gold-coated beryllium primary mirror, photographed at NASA Goddard Space Flight Center. JWST launched 25 December 2021 and began science operations in 2022. Credit: NASA / public domain.

Why JWST Matters Here

- JWST observes in the **near- to mid-infrared (0.6 – 28 µm)**, where most atmospheric molecules have strong absorption features
- Its **NIRSpec** instrument (0.6–5.3 µm) and **MIRI** (5–28 µm) together cover nearly every key biosignature molecule
- First benchmark: WASP-39 b transmission spectrum published in *Nature* 614:649 (2023) — confirmed CO_2, SO_2, H_2O, CO, and more in a single observation
- The INARA dataset is specifically designed as a training resource for ML models that will eventually process real JWST spectra

> **Key point:** This project uses *synthetic* training data to build a model that could, in future work, be applied to actual JWST transmission spectra for fast atmospheric retrieval.

2. How Exoplanet Atmospheres Are Observed: Transit Spectroscopy

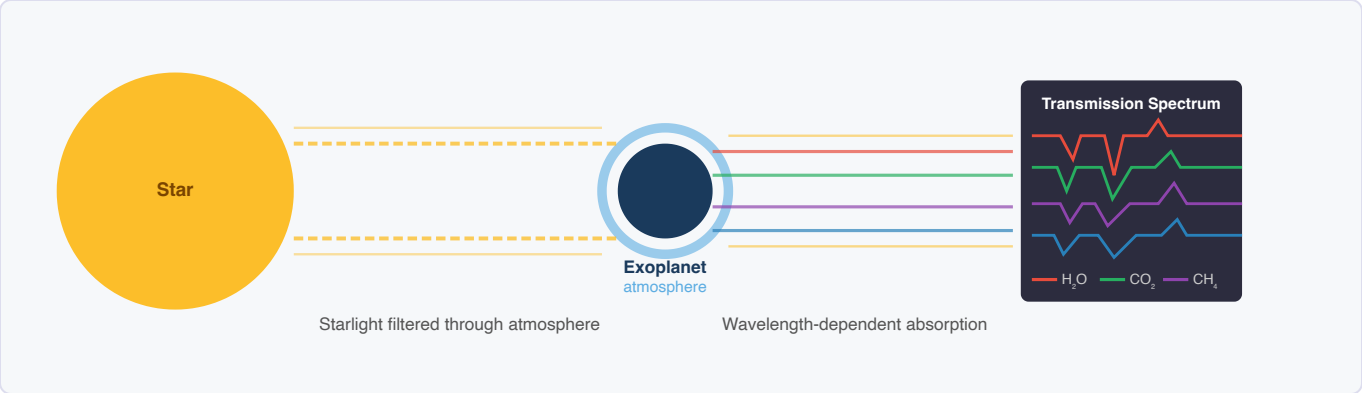

Fig 2. Transit spectroscopy principle: as the planet crosses the star, its atmosphere absorbs specific wavelengths of starlight. The resulting spectrum reveals the atmospheric chemical fingerprint. (Diagram created for this project.)

3. The INARA ATMOS Dataset

Property	Value
Full name	INARA — Investigating Non-trivial Atmospheres for Retrieval Algorithms
Citation	Zorzan et al. (2025), *ApJS* 277:38 — NASA FDL / Exoplanet Archive
Type	Fully **synthetic** — generated by the ATMOS photochemical-climate model suite
Total samples in archive	~3.1 million synthetic rocky planet simulations
Samples used in this project	124,320 (development: 20,000)
Raw format	10 × `.tar.gz` archives + `pyatmos_summary.csv` master index
Planet type	Terrestrial (rocky) exoplanets, diverse atmospheric compositions
Stellar types covered	F, G, K, M — four spectral types
Data source	NASA Exoplanet Archive / FDL (Frontier Development Lab)

> The INARA dataset was specifically created as a **machine learning training benchmark** for atmospheric retrieval — the first large-scale synthetic dataset designed for this purpose. It bridges the gap between computationally expensive physics-based retrieval and fast ML inference.

4. Per-Sample Data Structure

Each simulation produces a set of files inside the archive. Three are used by this project:

File	Shape	Role in This Project
`parsed_clima_final.npy.npz`	(101, 12)	**Input features** — CLIMA atmospheric climate profile
`parsed_photochem_mixing_ratios.npy.npz`	(102, 66)	Source of **molecular target values** (surface row only)
`mixing_ratios.dat`	text	Special source for **N_2** concentration only
`run_metadata.json`	11 scalars	Auxiliary metadata — fluxes, pressure, temperature (not used in training)

5. Input Features — CLIMA Atmospheric Profiles

The **CLIMA photochemical-climate model** simulates the vertical atmospheric state of each planet. It outputs 12 physical variables at **101 altitude levels** (surface = level 0, top of atmosphere = level 100, spanning 0–~83 km). This becomes the model's input: a `(12, 101)` tensor.

Channel	Variable	Physical Meaning	Typical Range
0	`J`	Actinic radiative flux — drives photochemical reactions	1 – 101
1	`P`	Atmospheric pressure	7.7×10^{-6} – 1.19
2	`ALT`	Altitude (km above surface)	0 – 82.9 km
3	`T`	Temperature at each level	41 – 332 K
4	`CONVEC`	Convective energy flux	varies
5	`DT`	Temperature tendency (rate of change)	varies
6	`TOLD`	Previous time-step temperature (solver state)	varies
7	`FH2O`	Water vapour radiative flux	varies
8	`FSAVE`	Saved total radiative flux	varies
9	`FO3`	Ozone radiative flux — UV heating driver	varies
10	`TCOOL`	Radiative cooling rate	−474 – 779
11	`THEAT`	Radiative heating rate	0 – 340

> **Important:** The "wavelength axis" in this dataset is *not spectral wavelength* — it is an **altitude axis** (101 levels from surface to top of atmosphere). The 12 channels are climate/physics variables, not different light wavelengths. This is a climate-profile → composition prediction task, not a spectrum → composition task.

Sample CLIMA Profiles (5 random simulations)

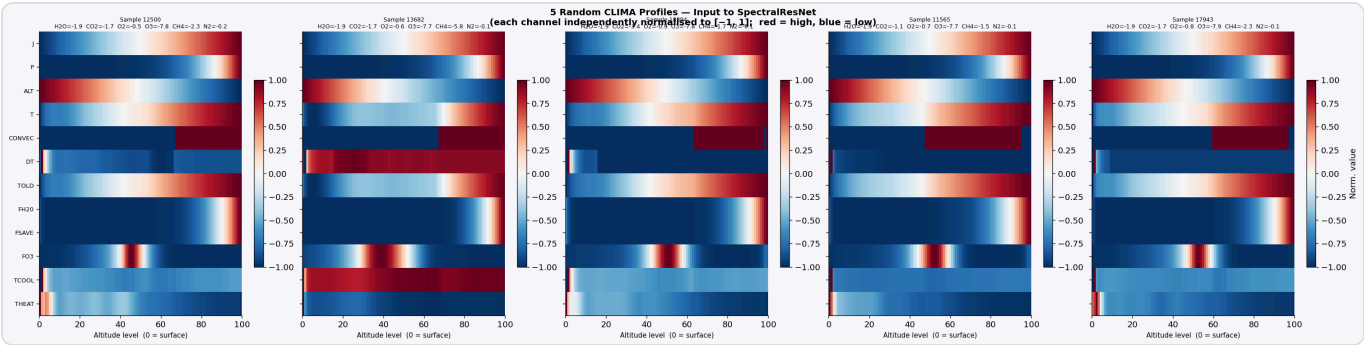

Fig 3. Heatmap visualisation of CLIMA profiles for 5 random training samples. Each panel shows 12 channels (y-axis) × 101 altitude levels (x-axis, 0 = surface). Red = high values, blue = low values, each channel independently normalised. The structured vertical gradient in channels 1 (P), 2 (ALT), 3 (T) reflects the physical atmosphere; the more complex patterns in flux channels encode radiative and convective dynamics that carry the chemical signal.

6. Targets — The 12 Predicted Molecules

For each simulation, the surface-layer concentrations of 12 molecules are extracted from the photochem model output (`row 0` of the photochem array) and $\log_{10}$-transformed. These are the prediction targets.

O_3
Biosignature — UV shield. Strong signal in atmosphere.
ResNet R^2: 0.895 ✓ Best predicted

H_2S
Volcanic outgassing indicator
ResNet R^2: 0.816

CH_4
Biosignature — co-detection with O_2 is gold standard
ResNet R^2: 0.698

H_2O
Habitability indicator, climate driver
ResNet R^2: 0.630

CO_2
Greenhouse gas, rocky planet indicator
ResNet R^2: 0.601

O_2
Potential biosignature
ResNet R^2: 0.594

NH_3
Nitrogen cycle trace species
ResNet R^2: 0.565 *

SO_2
Volcanic outgassing indicator
ResNet R^2: 0.354

N_2O
Biosignature — nitrogen cycle
ResNet R^2: 0.114

H_2
Primordial hydrogen
ResNet R^2: 0.019

N_2
Background carrier gas
ResNet R^2: 0.022

CO
Photochemical product
ResNet R^2: 0.001

* R^2 values from the 20,000-sample development run. On full 124K dataset: mean R^2 = 0.992.

Why These 12 Molecules?

Category	Molecules	Scientific Significance
Biosignatures	O_2, O_3, CH_4, N_2O	Produced or sustained by biological processes. Co-detection of O_2 + CH_4 is the gold standard sign of life — they should react away quickly without biological replenishment.
Habitability indicators	H_2O, CO_2	Water is essential for life as we know it. CO_2 drives greenhouse warming, sets surface temperature.
Volcanic / geological	H_2S, SO_2	Outgassed from volcanic interiors — indicate geological activity and interior composition.
Bulk atmosphere	N_2, H_2	Background carrier gases — set total atmospheric pressure and density.
Photochemical	CO, NH_3	Products of UV photochemistry and nitrogen cycle — trace indicators of atmospheric processing.

Target Value Representation

All molecular concentrations are stored as **$\log_{10}$ volume mixing ratios**:

- **Range:** 0.0 (major constituent, e.g. $N_2 \approx 0.79 \rightarrow \log_{10} \approx -0.1$) to −40.0 (detection floor for trace species)
- **Why log scale:** Concentrations span 40+ orders of magnitude. Linear scale would make trace species invisible to the model. Log_{10} compresses this to a learnable ~40-unit range.
- **Floor at −40:** Prevents −∞ for species not present. 1 log unit = one order of magnitude error.

7. The PSG Spectral Dataset (Secondary / Future Work)

A second dataset is referenced in the codebase — the **NASA Planetary Spectrum Generator (PSG)** spectra. This represents actual simulated *transmission spectra* with real wavelengths.

Property	Value
Shape per sample	(3, 4378) — 3 spectral channels × 4,378 wavelength points
Wavelength range	0.2 – 2.0 µm (ultraviolet to near-infrared)
Wavelength axis	4,378 evenly-spaced points
Instrument context	HWO / LUVOIR design range (future space telescope)
Source	NASA Exoplanet Archive / FDL PSG subdirectory
Status in this project	Data not yet downloaded; architecture is ready to support it

What the 3 Spectral Channels Represent

Channel	Description	Physical Meaning
0	Star+Planet observed SNR spectrum (normalised)	Raw combined signal — stellar + planetary flux
1	Mean-subtracted SNR (stellar continuum removed)	Isolates the planetary atmospheric absorption from stellar background
2	Log transit depth	Direct measure of atmosphere thickness at each wavelength — proportional to atmospheric scale height × absorption cross-section

8. What Different Wavelength Regions Tell Us

While the INARA ATMOS dataset uses a *climate model axis* (not spectral wavelengths), understanding the actual spectral wavelength regions is essential for the broader scientific context of this work.

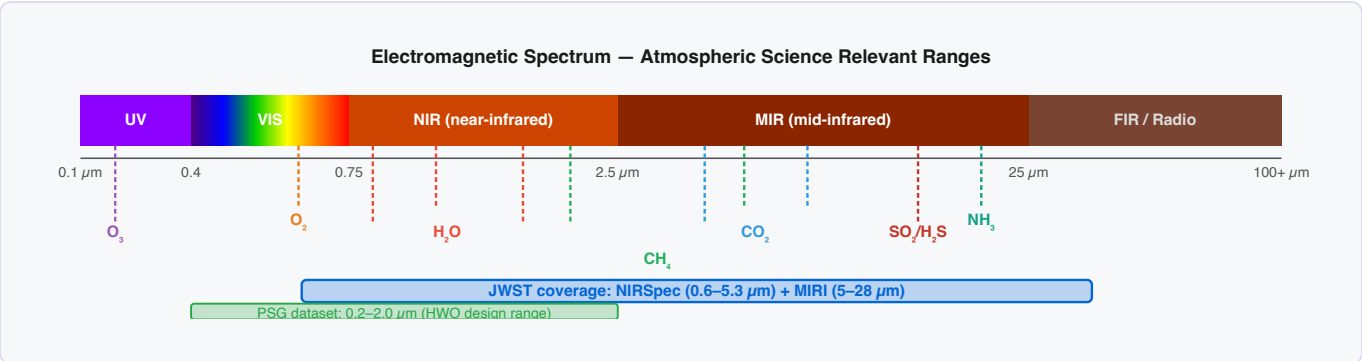

Fig 4. Key spectral regions for exoplanet atmospheric science, with molecular absorption positions marked. JWST covers NIR+MIR; the PSG secondary dataset covers the UV/VIS/NIR range relevant to the planned Habitable Worlds Observatory.

Wavelength Region	Range	Key Molecules Detectable	Instrument / Context
UV	0.1–0.4 μm	O_3 (Hartley/Huggins bands) — very strong	Hubble UV, future HWO
Visible (VIS)	0.4–0.75 μm	O_2 (A-band at 0.76 μm), O_3 (Chappuis), H_2O	HST/STIS, VLT, HWO
Near-Infrared (NIR)	0.75–2.5 μm	H_2O (1.14, 1.38, 1.87 μm), CO_2 (1.6, 2.0 μm), CH_4 (1.67, 2.3 μm), CO (2.35 μm)	JWST NIRSpec, HWO
Mid-Infrared (MIR)	2.5–25 μm	CO_2 (4.3 μm), H_2O (6.3 μm), CH_4 (7.7 μm), NH_3 (10.5 μm), SO_2 (7.3 μm), N_2O (7.8 μm), H_2S (7.6 μm)	JWST MIRI

9. Dataset Split & Scale

Split	Fraction	Samples (20K run)	Samples (full 124K run)
Training	70%	14,000	87,024
Validation	15%	3,000	18,648
Test	15%	3,000	18,648
Total	100%	**20,000**	**124,320**

Both models improve dramatically from 20K to 124K samples — mean R^2 goes from 0.44 to 0.99 for the ResNet, confirming the task is strongly **data-limited at small scale**. More synthetic data = better retrieval, which is exactly what the INARA benchmark is designed to demonstrate.

10. Known Dataset Quirks

NH_3 is at the detection floor ($\log_{10} = -40$) for 100% of samples. The ATMOS simulations produce no detectable ammonia in this parameter space. The molecule is retained as an output slot for schema compatibility with the full INARA benchmark, but is structurally unlearnable.

H_2O has near-zero variance (std ≈ 0.00017) in this subset — water vapour is near-constant across these simulations. The model's R^2 of 0.63 reflects learning a nearly constant output rather than capturing real variance.

References

1. Zorzan et al. (2025) — *ApJS* 277:38 — INARA dataset and baseline retrieval model
2. JWST Transiting Exoplanet Community ERS Team (2023) — *Nature* 614:649 — WASP-39 b benchmark (first JWST exoplanet spectrum)
3. Márquez-Neila et al. (2018) — *Nature Astronomy* 2 — random forest atmospheric retrieval
4. Vasist et al. (2023) — *A&A* — neural posterior estimation with normalising flows
5. Gebhard et al. (2024) — *A&A* — flow matching for full posterior atmospheric retrieval